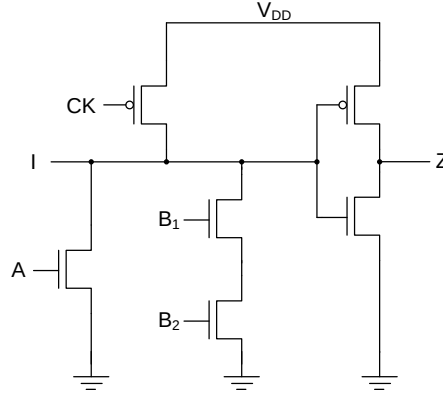


“数字集成电路设计” 2023-2024 第 1 学期 第 6 周作业

参考答案:

7.1 答:

(a)



(b)

用长沟道公式计算:

PMOS 管:

$$|V_{GS} - V_{T,p}| = 1.2 - 0.51 = 0.69$$

$$|V_{DS}| = V_{DD} - V_I$$

$\therefore V_I \leq 0.51$ 时, PMOS 在饱和区, $V_I \geq 0.51$ 时, PMOS 管线性

$$L_n = L_p = 40\text{nm} - 2 \times 10\text{nm} = 20\text{nm}$$

$$T_w = C_I \left[\int_{V_I=0V}^{V_I=0.51V} \frac{dV_I}{I_{D,sat}} + \int_{V_I=0.51V}^{V_I=0.6V} \frac{dV_I}{I_{D,linear}} \right]$$

$$I_{D,sat} = \frac{1}{2} \mu_p C_{ox,p} \frac{W_p}{L_p} (V_{SG} - |V_{th,p}|)^2 = 1.314\text{mA}$$

$$I_{D,linear} = \frac{1}{2} \mu_p C_{ox,p} \frac{W_p}{L_p} \left(2 \times (V_{SG} - |V_{th,p}|) \times (V_{DD} - V_I) - (V_{DD} - V_I)^2 \right) = 1.292\text{mA}$$

积分法: $T_w = 47.7041\text{ps}$

平均电流法: $T_w = 46.048\text{ps}$

用短沟道公式:

求 PMOS 饱和/线性临界电压

$$V_{dd} - V_I = \frac{(V_{SG} - |V_{T0,p}|) \times E_{C,p} L_p}{(V_{SG} - |V_{T0,p}|) + E_{C,p} L_p}$$

$$V_I = 0.7012\text{V}$$

$\therefore V_I \leq 0.7012$ 时, PMOS 在饱和区

$$T_W = C_I \left[\int_{V_I=0V}^{V_I=0.6V} \frac{dV_I}{I_{D,sat}} \right]$$

$$I_{D,sat} = \frac{1}{2} \mu_p C_{ox,p} \frac{W_p}{L_p} \frac{(V_{SG} - |V_{th,p}|)^2}{1 + \frac{(V_{SG} - |V_{th,p}|)^2}{E_{C,p} L_p}}$$

$$T_W = 63.16\text{ps}$$

7.6 答：

与非门实现： $F = AB + AC + BC = \overline{(\overline{AB}) \cdot (\overline{AC}) \cdot (\overline{BC})}$

或非门实现： $F = (A+B)(A+C)(B+C) = \overline{(\overline{A+B}) + (\overline{A+C}) + (\overline{B+C})}$

7.7 答：

第一种情况：两个输入同时跳变

等效宽长比为：

$$(W/L)_{eq,n} = 2(W/L)_n = 2, \quad k_{n,eq} = k_n' (W/L)_{eq,n} = 196.4 \mu A/V^2$$

$$(W/L)_{eq,p} = \frac{1}{2}(W/L)_p = 2, \quad k_{p,eq} = k_p' (W/L)_{eq,p} = 92 \mu A/V^2$$

$$V_{OL} = 0V$$

$$V_{OH} = V_{DD} = 1.2V$$

计算 V_{IL}

$$V_{out} = \frac{1}{2} \left\{ \left(\frac{k_n}{k_p} + 1 \right) V_{IL} - \frac{k_n}{k_p} V_{T0,n} - V_{T0,p} + V_{DD} \right\}$$

$$V_{out} = 1.567V_{IL} + 0.289$$

$$\frac{k_n}{2} (V_{IL} - V_{T0,n})^2 = k_p \left(V_{IL} - V_{DD} - V_{T0,p} - \frac{V_{out} - V_{DD}}{2} \right) (V_{out} - V_{DD})$$

$$V_{IL} = 0.2072V \text{ 或 } 0.5708V$$

对应的 $V_{OUT} = 0.614V$ 或 $1.184V$

因为此时 NMOS 处于饱和区, 需要 $V_{IL} \geq V_{TH}$

舍去 0.2072V 这个解（此时 nmos 截止）

所以： $V_{IL} = 0.5708V$

计算 V_{IH}

$$V_{out} = \frac{1}{2} \left\{ \left(1 + \frac{k_p}{k_n} \right) V_{IH} - V_{T0,n} - \frac{k_p}{k_n} (V_{DD} + V_{T0,p}) \right\}$$

$$V_{out} = 0.734V_{IH} - 0.427$$

$$k_n \left(V_{IH} - V_{T0,n} - \frac{V_{out}}{2} \right) V_{out} = \frac{k_p}{2} (V_{IH} - V_{DD} - V_{T0,p})^2$$

$$V_{IH} = 0.6102V$$

$$NM_L = 0.5708 - 0 = 0.5708V$$

$$NM_H = 1.2 - 0.6102 = 0.5898V$$

第二种情况：只有一个输入跳变（不考虑体效应）

等效宽长比为：

$$(W/L)_{eq,n} = (W/L)_n = 1$$

$$(W/L)_{eq,p} = \frac{1}{2} (W/L)_p = 2$$

$$V_{OL} = 0V$$

$$V_{OH} = V_{DD} = 1.2V$$

计算 V_{IL} $V_{IL} = 0.5885V$

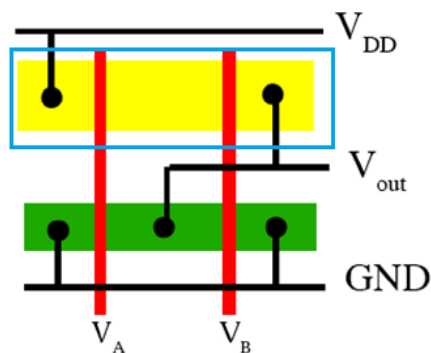
计算 V_{IH} $V_{IH} = 0.6283V$

$$NM_L = 0.5885 - 0 = 0.5885V$$

$$NM_H = 1.2 - 0.6283 = 0.5717V$$

补充题：

1、答：



2、答：二输入与非门，电路图（略）